PMC: Build-Time Per-Modality Centroid Correction for Cross-Modal Binary-Quantized Retrieval

Anonymous Author(s)

Abstract

Approximate nearest neighbor search is a core operator in large-scale retrieval, where vector quantization is widely used to reduce memory cost. Binary quantization methods encode each dimension as a single bit, but they assume queries and database vectors share a common centroid. This assumption fails in cross-modal retrieval, where each modality clusters around a different centroid, producing a modality gap that corrupts the centroid's dual role as inverted-file routing pivot and sign-bit decision boundary. We propose Per-Modality Centroid Correction (PMC), which shifts database vectors toward the query centroid at build time, rewriting routing and code boundaries at the source rather than merely adjusting queries at serving time. A selective variant confirms that concentrated gap energy in a few dimensions drives most recall loss. PMC adds no index memory or serving overhead; at α=1, the correction is fully absorbed into the index with zero query-time cost. Experiments on five cross-modal benchmarks, three encoders, and four binary-quantized index types show gains over uncorrected and query-shifted baselines that scale with modality-gap magnitude, up to a 400M-vector deployment. Code: https://anonymous.4open.science/r/PMC-CIKM.

CCS Concepts

• **Information systems** → **Multimedia and multimodal retrieval**; *Search engine indexing*.

Keywords

cross-modal retrieval, modality gap, approximate nearest neighbor search, binary quantization

ACM Reference Format:

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1 Introduction

Multimodal foundation models such as CLIP [23] and ImageBind [9] map heterogeneous signals (images, text, audio) into a shared vector space for cross-modal vector similarity search [15]. At deployment scale, storage becomes the bottleneck: high-dimensional `float32` embeddings can require terabytes of RAM, motivating approximate nearest neighbor (ANN) indexes that compress to a few bytes per vector [7, 14].

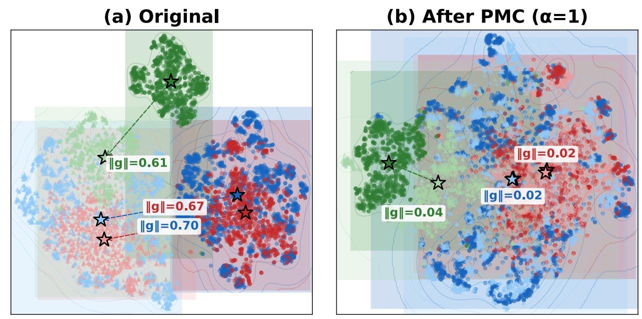

Figure 1: t-SNE of ImageBind embeddings. (a) Each modality forms a separate cluster. (b) After PMC (α=1), paired modalities overlap. Blue = MSCOCO, red = Flickr30K, green = Clotho; dark = DB, light = query.

Binary quantization (BQ) is the extreme compression point: RaBitQ [7] reaches 32× compression by encoding each dimension as one sign bit relative to the dataset centroid. Its production adoption in Apache Lucene BBQ and extension to multi-bit Extended-RaBitQ [6] show BQ indexing is now practically deployed.

These methods rely on a shared-centroid residual premise: queries and database vectors are drawn from the same distribution and share a common centroid. Cross-modal retrieval violates this assumption—queries from one modality cluster around a different centroid than the database vectors.

This *modality gap* [21] is particularly harmful to BQ because the encoding $\text{sign}(x_i - c_i)$ has *zero error margin*: the decision boundary passes exactly through the centroid, and any systematic displacement causes structured bit corruption that scales with gap magnitude (Figure 1); an oracle test confirms this: same-modality queries ($\|\mathbf{g}\|$=0) reach R@100=0.84 under RaBitQ, versus 0.57 for cross-modal queries ($\|\mathbf{g}\|$=0.82). Prior work addresses cross-modal retrieval through graph augmentation [2, 13], encoder fine-tuning [5], or embedding-space calibration [19], but none targets compressed ANN indexes where centroid misalignment corrupts both Inverted File (IVF) routing and binary codes.

In this paper, we first analyze how the modality gap degrades BQ ANN indexes at two coupled stages—IVF routing and binary code formation—and show that dimensions with concentrated gap energy are most vulnerable to boundary crossing; a selective correction test confirms that correcting only the highest-gap dimensions recovers most of the gain. Based on this analysis, we propose Per-Modality Centroid Correction (PMC), a one-time build-time alignment that shifts database vectors toward the query centroid before index construction. Unlike query-only mean shift, which leaves IVF centroids and quantized codes misaligned, PMC exploits the centroid's dual role as IVF routing pivot and sign-bit decision

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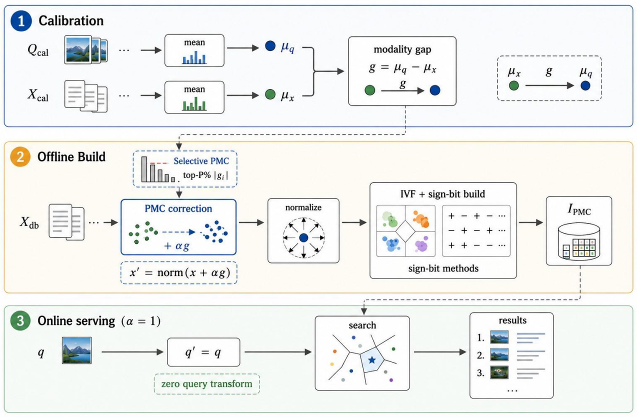

Figure 2: PMC workflow: calibration estimates μ_q, μ_x, and g=$\mu_q - \mu_x$; offline build shifts/normalizes database vectors before BQ index construction; at α=1, online serving uses q′=q and searches $\mathcal{I}_{\text{PMC}}$.

boundary, rewriting both at the source with no serving-time query transform and no additional index memory (Figure 2). We validate PMC across five cross-modal benchmarks including 407M-scale LAION, three multimodal encoders, and four BQ index types. Our contributions:

- We show that the modality gap corrupts both IVF routing and binary codes, with recall loss scaling with gap magnitude.
- We discover that this corruption concentrates in a few high-$|g_i|$ dimensions; a selective PMC variant correcting only those recovers most recall.
- We propose PMC, a zero-cost build-time correction (α=1) that realigns routing and codes via the centroid's dual role; ablation confirms DB-side over query-side correction.

2 Related Work

Vector quantization for ANN. Classical ANN compression methods such as Product Quantization (PQ)/OPQ [8, 14] quantize vectors by subspace codebooks, while newer systems such as ScaNN/SOAR/QAQ [12, 28, 33] improve scoring, routing, or query-aware ranking through anisotropic quantization and query-dependent computation. In the 1-bit regime, RaBitQ [7] provides distortion bounds for BQ, Extended-RaBitQ [6] generalizes this line to multi-bit settings, and later variants integrate quantization with graph structures [11]. Binary hashing methods such as ITQ [10] similarly rely on distributional alignment. Across these lines, the standard assumption is that query and database statistics are in-distribution and approximately centroid-aligned.

Cross-modal and OOD retrieval. Cross-modal and OOD retrieval methods usually intervene outside compressed code formation. OOD-DiskANN [13, 27] augments graph search for distribution shift; RoarGraph/DEG [2, 31] modify graph connectivity; DeDrift [1] and TCR [20] use test-time adaptation or retraining-style updates to counter drift. They are effective but usually assume uncompressed vectors, additional graph structures, or online serving updates. PMC is complementary: one offline centroid correction before index construction, with no added index-time memory or required retraining.

Table 1: BQ methods (R@100, Vanilla / PMC) on MSCOCO (CLIP-B/32) and AudioCaps (IB). RotatedBinary = rotated BinaryFlat (BBQ-style).

	MSCOCO		AudioCaps	
Method	t⟶i	i⟶t	t⟶a	a⟶t
BinaryFlat	.51 / **.57**	.46 / **.57**	.63 / **.66**	.51 / **.64**
BinaryIVF	.51 / **.57**	.47 / **.57**	.59 / **.61**	.50 / **.63**
RotatedBinary	.29 / **.58**	.21 / **.57**	.65 / **.70**	.63 / **.69**
RaBitQ	.58 / **.64**	.52 / **.61**	**.75** / **.75**	.67 / **.75**

Algorithm 1: PMC Index Construction

Require: Database $\mathcal{X}$ (modality b), query sample Q (modality a), interpolation α
Ensure: IVF-based quantized index $\mathcal{I}$ and gap g

```
μ_q ← mean(Q);   μ_x ← mean(X)
g ← μ_q − μ_x
for each x ∈ X do
    x′ ← normalize(x + α g)
end for
I ← BuildIndex({x′})                ▷ RaBitQ, BBQ, BinaryFlat
return I, g
```

Modality gap. Mind-the-Gap [21] shows that cross-modal embedding spaces often exhibit stable centroid offsets between modalities. Follow-up analyses attribute this behavior to training dynamics [25], characterize its geometry [18], and study mean-subtraction effects [34]. Other work either preserves the gap [24], reduces it through encoder retraining [5], or exploits gap-aware calibration for *uncompressed* ranking [19]; see also the recent survey [30]. No prior work addresses how the modality gap degrades *compressed* ANN indexes, where centroid misalignment corrupts both IVF routing and quantized codes; this is our focus.

Closest to our setting, gap-aware calibration for mixed-modality ranking [19] adjusts scores to compensate for the centroid offset, but operates on uncompressed vectors and does not enter code formation; PMC targets the compressed index itself, rewriting IVF routing and binary codes at the source before quantization.

3 Method

3.1 Problem Setup

Let ϕ be a frozen multimodal encoder mapping inputs from modalities a and b into $\mathbb{R}^d$. The database $\mathcal{X} = \{\phi(x_i)\}$ consists of modality-b embeddings; queries $q = \phi(q_j)$ come from modality a. We denote the per-modality means $\boldsymbol{\mu}_q = \mathbb{E}[\phi(q)]$ and $\boldsymbol{\mu}_x = \mathbb{E}[\phi(x)]$, and the *modality gap* $\mathbf{g} = \boldsymbol{\mu}_q - \boldsymbol{\mu}_x$. The goal is to build a compressed ANN index over $\mathcal{X}$ that answers cross-modal queries with high recall under a fixed memory budget.

3.2 BQ Vulnerability to Modality Gap

BQ encodes each dimension as $\hat{x}_i = \text{sign}(x_i - c_i)$ relative to centroid c, as in RaBitQ [7] and BBQ (Apache Lucene). With one bit, the decision boundary passes exactly through c with *zero error margin.* Let $\delta_i = x_i - c_i$ be the residual under $c = \boldsymbol{\mu}_x$ (so $b_i = \text{sign}\,\delta_i$); shifting to $c^* = \boldsymbol{\mu}_x + \alpha\mathbf{g}$ flips dimension i's sign iff

$$\delta_i(\delta_i - \alpha g_i) < 0, \quad \text{i.e.,} \quad 0 < \frac{\delta_i}{\alpha g_i} < 1\,. \tag{1}$$

Under smooth symmetric residual densities p_i near the decision boundary, a first-order expansion yields the expected number of

Table 2: PMC at α=1 for IVFRaBitQFastScan (Vanilla/MeanShift/PMC; relative R@100 gains Δ_{VP}=Vanilla→PMC, Δ_{MP}=MeanShift→PMC).

Dataset	Enc.	$\lVert g \rVert$	$q \to db$				$db \to q$			
			R@10	R@100	Δ_{VP}	Δ_{MP}	R@10	R@100	Δ_{VP}	Δ_{MP}
MSCOCO	CLIP	.82	.39 / .37 / **.46**	.57 / .53 / **.63**	+10%	+17%	.28 / .30 / **.38**	.49 / .49 / **.60**	+22%	+22%
MSCOCO	CL-L	.82	.36 / .37 / **.48**	.54 / .53 / **.65**	+20%	+22%	.26 / .34 / **.44**	.46 / .50 / **.63**	+37%	+25%
MSCOCO	IB	.70	.56 / .51 / **.63**	.67 / .62 / **.75**	+12%	+21%	.57 / .54 / **.64**	.70 / .66 / **.75**	+7%	+14%
Flickr30K	CL-L	.77	.29 / .25 / **.37**	.39 / .34 / **.48**	+23%	+41%	.21 / .27 / **.38**	.31 / .35 / **.46**	+48%	+31%
Clotho	IB	.61	.59 / .44 / **.59**	.72 / .60 / **.73**	+2%	+22%	.48 / .35 / **.53**	.62 / .51 / **.68**	+11%	+34%
AudioCaps	IB	.61	.36 / .45 / **.39**	.75 / .78 / **.79**	+5%	+0%	.43 / .47 / **.45**	.82 / .82 / **.83**	+1%	+1%
LAION-400M	CLIP	.72	.068 / .037 / **.082**	.095 / .072 / **.137**	+44%	+90%	.031 / .028 / **.046**	.058 / .042 / **.072**	+24%	+71%

$q \to db$: text-to-image/audio; $db \to q$: reverse. Each cell reports Vanilla / MeanShift / **PMC**. CLIP = CLIP ViT-B/32; CL-L = CLIP-L; IB = ImageBind. MeanShift = query-only mean shift. LAION: n_{list}=20K, n_{probe}=256; single-thread CPU (i7-12700F).

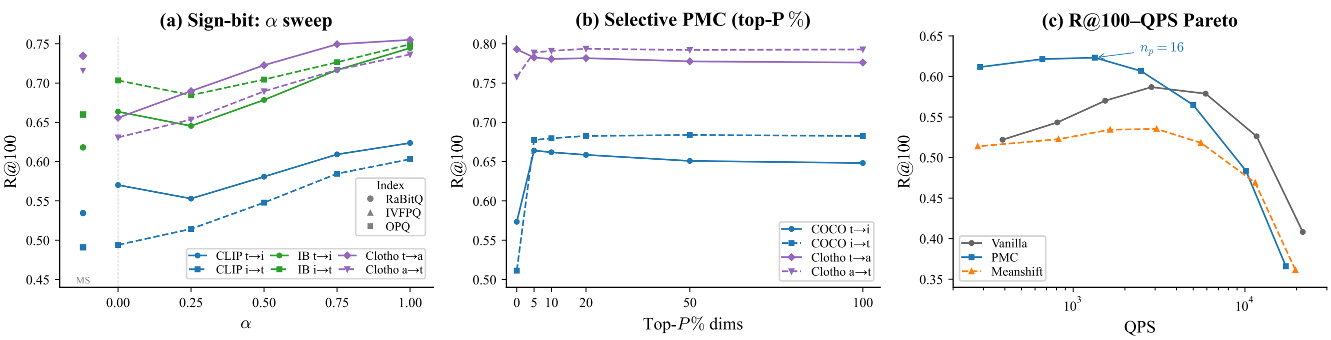

Figure 3: (a) BQ R@100 vs. shift strength α on MSCOCO+Clotho (α=0: query-only mean shift, α=1: full DB-side PMC): α=1 is best or near-best, justifying the default. (b) Selective PMC: on concentrated-gap MSCOCO, top-P=5% already reaches peak R@100; diffuse gaps require broader correction. (c) R@100–QPS Pareto: PMC dominates vanilla across throughput levels.

flips:

$$\mathbb{E}[F] \;\approx\; \alpha \sum_i p_i(0)\,|g_i|\,, \tag{2}$$

a density-weighted ℓ_1 norm of the gap. Eq. (1) shows that flip risk in dimension i grows with $|\alpha g_i|/|\delta_i|$: larger gap components produce wider flip intervals. When gap energy is concentrated—in CLIP-L on MSCOCO, top 10% of dimensions carry ≈90% of $\lVert \mathbf{g} \rVert^2$—corruption is *structured* rather than random, biasing Hamming distance estimates systematically. The same offset misaligns IVF cluster assignments, compounding routing errors with code corruption. Eq. (2) formalizes this scaling, explaining the gap-proportional gains in Table 2 and why top-5% selective PMC already reaches peak recall (Figure 3b). Multi-bit quantization (IVFPQ [14], OPQ [8]) partially absorbs displacement through wider error margins; BQ is maximally vulnerable and is the focus of our main claims (multi-bit: Section 4.5). PMC applies to all sign$(\cdot - c)$ methods (BinaryFlat, BinaryIVF, BBQ, RaBitQ) and analogous schemes such as LSH [29].

3.3 PMC Correction

PMC applies a one-time centroid alignment *before* index construction. Given a training sample of queries from modality a and the database from modality b, we compute the gap $\mathbf{g}$ and shift database vectors toward the query centroid:

$$\mathbf{x}' = \frac{\mathbf{x} + \alpha\,\mathbf{g}}{\lVert \mathbf{x} + \alpha\,\mathbf{g} \rVert}\,, \qquad \mathbf{q}' = \frac{\mathbf{q} - (1-\alpha)\,\mathbf{g}}{\lVert \mathbf{q} - (1-\alpha)\,\mathbf{g} \rVert} \tag{3}$$

where $\alpha \in [0, 1]$ interpolates between full query-side correction (α=0, equivalent to mean shift) and full database-side alignment (α=1). The BQ index is then built on $\{\mathbf{x}'\}$; in our main setting α=1 (full database-side alignment), query-time transformation is identity ($\mathbf{q}'$=$\mathbf{q}$), so serving incurs no extra cost.

Effectiveness of $\boldsymbol{\alpha}$=1. Per-vector renormalization after the shift preserves unit-norm structure; the α-sweep in Figure 3a confirms α=1 as best or near-best across all tested settings.

Selective PMC as a mechanism test. We restrict correction to top-P% dimensions ranked by $|g_i|$:

$$g_{\mathrm{sel},i} = \begin{cases} g_i & \text{if } |g_i| \geq |g|_{(P)} \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

$$\mathbf{x}'_{\mathrm{sel}} = \frac{\mathbf{x} + \alpha\,\mathbf{g}_{\mathrm{sel}}}{\lVert \mathbf{x} + \alpha\,\mathbf{g}_{\mathrm{sel}} \rVert} \tag{5}$$

where $|g|_{(P)}$ is the top-P% magnitude threshold on $|g_i|$. The cumulative energy captured by $\mathbf{g}_{\mathrm{sel}}$ is

$$\mathcal{E}(P) = \frac{\sum_{i \in \mathcal{S}_P} g_i^2}{\lVert \mathbf{g} \rVert^2}\,, \quad \mathcal{S}_P = \left\{ i : |g_i| \geq |g|_{(P)} \right\}. \tag{6}$$

Since flip risk concentrates in high-$|g_i|$ dimensions (Eq. (1)), correcting only the highest-energy ones suffices: for CLIP the top 10% by $|g_i|$ carry 86–92% of total gap energy across all tested datasets, so P=5% already recovers peak recall; ImageBind's diffuse gap (≈72%) requires full-vector correction.

4 Experiments

4.1 Setup

We evaluate three frozen backbones: CLIP ViT-B/32 (d=512) [23], CLIP ViT-L/14 (d=768), and ImageBind (d=1024) [9]. Datasets span

Table 3: Mechanism/control checks at α=1 (original exact-IP GT). J@100 = top-100 Jaccard overlap vs. exact GT.

Direction	Flip%	J@100	BinaryFlat R@100	n=25 cos
MSCOCO t→i	16.14	0.5852	0.5073 → 0.5697	0.9860
MSCOCO i→t	16.00	0.5422	0.4579 → 0.5675	0.9863
AudioCaps t→a	14.20	0.8016	0.6298 → 0.6597	0.9653
AudioCaps a→t	17.75	0.8036	0.5143 → 0.6442	0.9546

Table 4: MSCOCO/CLIP-B/32 checks: calibration sensitivity, component ablation, and IVF-RaBitQ (IVFRaBitQFastScan) controls. Rand/Shuf are mean±std over 5 seeds.

Calibration sensitivity

n_{calib}	t→i R@100	cos	i→t R@100	cos
25	.7294	.9860	.6938	.9863
100	.7296	.9965	.6950	.9964
400	.7296	.9992	.6947	.9992

Component ablation

Index	Dir.	Van.	Q-only	DB-only	Both
BinaryFlat	t→i	.507	.569	.570	.570
BinaryFlat	i→t	.458	.571	.568	.569
IVF-RaBitQ	t→i	.578	.541	**.637**	.599
IVF-RaBitQ	i→t	.515	.495	**.608**	.554

IVF-RaBitQ controls

Dir.	Van.	DB-PMC	Rand	Shuf	Flip	NoNorm
t→i	.578	**.637**	.562±.009	.552±.006	.514	.555
i→t	.515	**.608**	.564±.009	.559±.009	.491	.520

image–text and audio–text retrieval: MSCOCO 5K [22], Flickr30K (n=31K) [32], AudioCaps [17] (884 clips, 4,415 captions; text→audio retrieves over audio DB), Clotho v2 (n=5,926) [4], and at large scale LAION-400M (407M vectors across 410 shards; exact brute-force inner-product (IP) ground truth over 10K random queries per direction, computed per shard with exact `IndexFlatIP` search and merged into the global top-100, identical to a single full-corpus scan) [26]. Indexes are built with FAISS [3, 16]: IVFRaBitQFastScan (1-bit, 33–136 B/vec) and IVFPQ/OPQ [8, 14] (m=16, 64 B/vec); default n_{list}=64, n_{probe}=16; LAION uses n_{list}=20K, 30.4 GB, 72 B/vec (28×); PMC leaves bytes per vector identical to vanilla; BQ baselines include BinaryFlat, BinaryIVF, and rotated BinaryFlat (BBQ-style). We report R@10 and R@100 against exact-IP ground truth on original (unshifted) embeddings—PMC modifies only the index-internal representation (analogous to OPQ rotation), so gains reflect better approximation, not redefined relevance—and QPS (queries per second) as the single-threaded median over 5 runs. All experiments run on a single Intel Core i7-12700F, 32 GB RAM, CPU-only.

4.2 Main Results

PMC improves RaBitQ recall across nearly all tested configurations (Table 2). Gains scale with gap magnitude: Flickr30K i→t reaches +48% and MSCOCO CLIP-L i→t +37% at high gap. R@10 gains are usually smaller than R@100, consistent with IVF routing effects on larger candidate sets. Query-only mean shift can hurt—it leaves IVF centroids misaligned, degrading routing—while PMC improves both directions (Table 2). Across all 16 BQ method×direction configurations, PMC improves or matches R@100 without exception (Table 1). Query throughput is unchanged: PMC achieves within 1% of vanilla QPS across all configurations (Figure 3c).

$\|\mathbf{g}\|$ **as predictor.** The ratio condition (1) implies more flips at higher gap, matching the Flickr30K/MSCOCO vs. AudioCaps contrast: AudioCaps shows minimal gain ($\Delta_{\text{VP}}\leq$5%) because its gap

Table 5: PMC generality on IVFPQ and OPQ across encoders.

Dataset	Enc.	Dir.	IVFPQ		OPQ	
			R@100	Δ	R@100	Δ
MSCOCO	CLIP	t→i	.54 / **.63**	+16%	.61 / **.66**	+8%
		i→t	.50 / **.64**	+30%	**.67** / **.67**	−1%
	CL-L	t→i	.43 / **.63**	+46%	.56 / **.67**	+20%
		i→t	.35 / **.58**	+69%	.60 / **.67**	+12%
	IB	t→i	.59 / **.70**	+18%	.69 / **.76**	+11%
		i→t	.59 / **.68**	+16%	.72 / **.78**	+7%
Flickr	CL-L	t→i	.51 / **.54**	+5%	**.55** / .54	−1%
		i→t	.43 / **.47**	+9%	.48 / **.50**	+5%

($\|\mathbf{g}\|$=0.61) is small relative to intra-class variance. At deployment scale, LAION-400M confirms the pattern in both retrieval directions at 407M vectors with identical index memory (Table 2).

4.3 Sign-Bit Corruption Analysis

We validate the predicted sign-bit corruption mechanism through gap-magnitude correlation, bit-flip statistics, and selective correction. R@100 gains grow with $\|\mathbf{g}\|$: high-gap settings show the largest improvements while low-gap datasets improve modestly (Table 2). PMC induces 14–18% sign-bit flips with J@100 of 0.54–0.80 (Table 3), confirming the predicted mechanism; gap direction is stable from n=25 samples (cosine $\geq$ 0.95). Selective PMC recovers gains from high-$|g_i|$ dimensions first: on MSCOCO/CLIP, $\mathcal{E}(5\%)\approx$87% already recovers peak R@100 (Figure 3b); ImageBind's diffuse gap ($\mathcal{E}(10\%)\approx$72%) requires broader correction.

4.4 Ablation Study

Calibration is stable from n_{calib}=25 (cosine ≈ 0.986), making PMC practical with few cross-modal samples. BinaryFlat is insensitive to shift direction, but IVF-RaBitQ strongly prefers DB-only correction (.637/.608) over Q-only (.541/.495) or both sides (.599/.554)—shifting both sides re-displaces the routing pivot after code boundaries improve, confirming that the centroid must be corrected at the source where it simultaneously serves as routing pivot and code boundary. Four controls (random, shuffled, sign-flipped, un-normalized shift) all trail DB-PMC (Table 4), ruling out arbitrary perturbation.

4.5 Multi-Bit Generality

PMC generalizes beyond 1-bit: it improves all shown IVFPQ settings and most OPQ settings (audio omitted; n_{db} too small for codebook training), with the gap-proportional trend persisting (Table 5). OPQ's learned rotation partially absorbs distributional shift, yielding smaller but complementary PMC gains; IVFPQ lacks this compensation and benefits more.

5 Conclusion

PMC addresses cross-modal centroid mismatch in compressed ANN through a build-time database-side correction: by exploiting the centroid's dual role as IVF routing pivot and sign-bit decision boundary, it rewrites these baked-in index structures rather than redefining relevance or shifting only queries. Under fixed exact-IP ground truth, PMC yields recall gains that scale with modality-gap magnitude, with no extra index memory and, at α=1, no serving-time query transform. Future directions include drift-adaptive α for streaming settings, PMC-style correction for graph-based ANN, and billion-scale deployment.

GenAI Usage Disclosure

Generative AI tools (Claude) were used for code scaffolding and LaTeX formatting. All experimental design, results, analyses, and scientific claims were independently produced and verified by the authors.

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